5	(a)	$E = I(R + r)$ $9.0 = 0.450(18 + r)$ $r = 2.0\ \Omega$	M1 A1
	(b)	$Q = It$ $= 0.450 \times 3600$ $= 1.6 \times 10^3\,\text{C}$	M1 A1
	(c)	$E_{total} = IVt$ $= 0.450 \times 9.0 \times 3600$ $= 14.6 \times 10^3\,\text{J}$	M1 A1
	(d)	$P = I^2R$ $= (0.450)^2 \times 18.0$ $= 3.65\ \text{W}$	M1 A1
	(e)	New effective resistance $= \dfrac{V}{I} = \dfrac{9.0}{0.225} = 40.0\ \Omega$ Replacement resistance = 40.0 – 2.0 = 38.0 Ω	M1 A1